8.7 Activities: ready for the brugklas

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1. Thermometer and lift

You need: a strip of paper about 40 cm long, a pen, a counter or coin.

Draw a vertical number line from −10 at the bottom to +10 at the top, with 0 in the middle. Call it a thermometer (or a lift in a building with a car park underneath: floor −1, −2, −3).

Call out moves: "It's 3 degrees. It gets 5 degrees colder." He moves the counter and reads the answer: −2. "The lift is at floor −2 and goes up 6 floors." Floor 4. Then let him call out moves for you.

Look up today's temperature in a cold place (Oslo, Moscow, Antarctica) and a warm one. How many degrees difference?

Talk about it: "When did you last see a negative number in real life?" (Freezer, winter weather, a bank balance, below sea level: much of the Netherlands is!)

2. Square tiles

You need: squared paper, coloured pencils, or Lego/tiles.

Build squares: 1 by 1, 2 by 2, 3 by 3 … up to 10 by 10. Count the tiles each time and write: $1^2 = 1$, $2^2 = 4$, $3^2 = 9$ … Then reverse: "I have 49 tiles in a square. How long is a side?" That's the square root: $\sqrt{49} = 7$.

Try a number that doesn't make a square, like 50 tiles. Can you make a perfect square? (No: $7 \times 7 = 49$ is one short; $8 \times 8 = 64$ is too many. So $\sqrt{50}$ is between 7 and 8.)

Talk about it: "Why do you think it's called 'squared'?"

3. Guess my rule (number machine)

You need: paper, a pencil.

One person secretly thinks of a rule, like "× 3, then + 2". The other says an input number; the rule-maker says the output. Fill in a table as you go:

in	1	2	5	10
out	5	8	17	32

Try to guess the rule from the table. Then write it with a letter: $out = 3 \times in + 2$, or $y = 3x + 2$. Start with one-step rules (+ 4, × 5) and build up.

Talk about it: "Which input numbers are most useful for guessing the rule?" (0, 1 and 10 are very revealing.)

4. Bracket battle

You need: dice (2 or 3), paper.

Roll three dice, for example 2, 3 and 5. Each player makes the **biggest** number they can using the three numbers, any of +, −, × and brackets. For example $(2 + 3) \times 5 = 25$, or $2 \times 3 \times 5 = 30$. Then try to make the **smallest** number, or a target number like 11. Explain the order the sum is done in.

Talk about it: “How do brackets change the answer? Can you find a sum where brackets make no difference?”

5. Brugklas role play

You need: two chairs, a “textbook” (any book), the help sentences from the README.

He’s the pupil, you’re the teacher (then swap). Act out short scenes:

- He doesn’t understand an exercise in the lesson and asks for help.
- He forgot his homework. What does he say?
- He got a 4,8 on a test. He goes to ask the teacher what to practise.
- He wants to tell his mentor that maths makes him nervous.

Use the Dutch sentences. Make it a bit funny: an over-strict teacher, a very kind one. The point is that saying the words feels normal.

Talk about it: “Which of those would be hardest to do for real? What would make it easier?”

6. Visit or look around

You need: the secondary school’s website or open day (*open dag*).

Look at the school’s maths pages, the timetable, the book list. If there’s an open day or welcome afternoon (*kennismakingsmiddag*), visit the maths classroom. Seeing the place shrinks the unknown.

Talk about it: “What are you most looking forward to? What are you unsure about?”

Conversation prompts for any day

- “It’s −4 °C now and +6 °C this afternoon. How much warmer will it get?”
- “What’s 9 squared? And the square root of 81?”
- “If a = 3, what is 2a + 1?”
- “What would you say to a teacher if you didn’t understand something?”
- “Who could you go to at your new school if something isn’t going well?”